MONASH University

ETW3482

Week Six Tutorial



This week's tutorial will have **two sessions**

1. **Quizzes for REVISION**
2. **Demo session: Building a Pipeline with various Logistic Regression Models**



MONASH University

Do you dare to take the challenge?



MONASH University

Do not edit
How to change the design



What are the assumptions of simple linear regression? (Choose two)

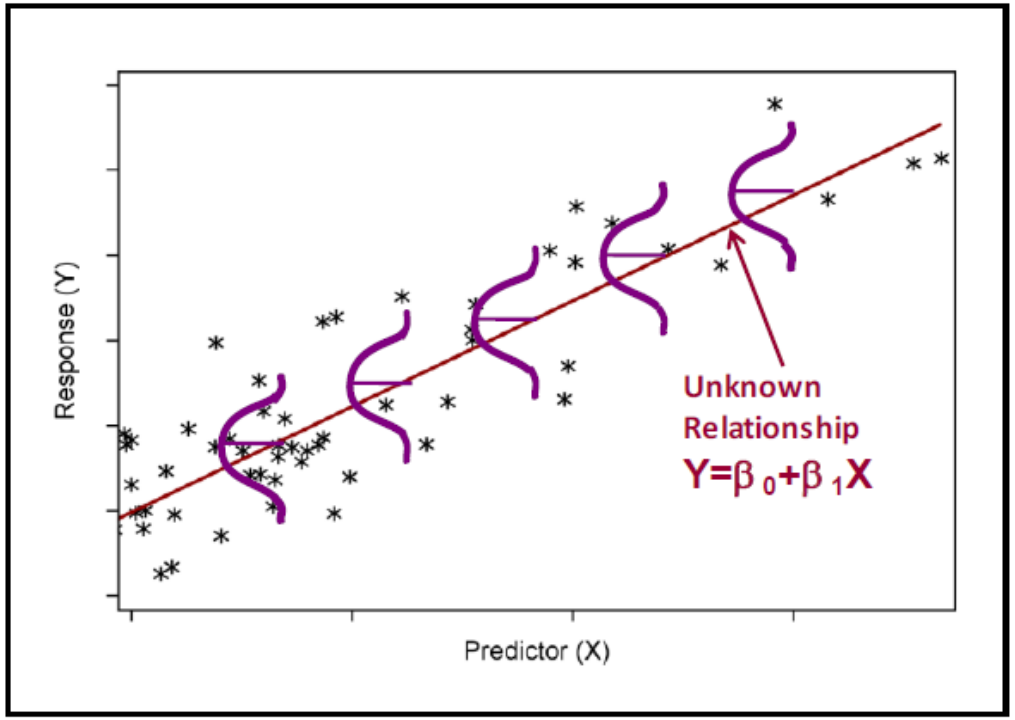
ⓘ The Slido app must be installed on every computer you're presenting from

slido

Assumptions of simple linear regression

The following are the assumptions of simple linear regression.

- The mean of the target variable is linearly related to the value of the input variable.
- The error is normally distributed.
- The error has constant variance across the range of the input variable.
- The observations are independent.



Do not edit
How to change the design



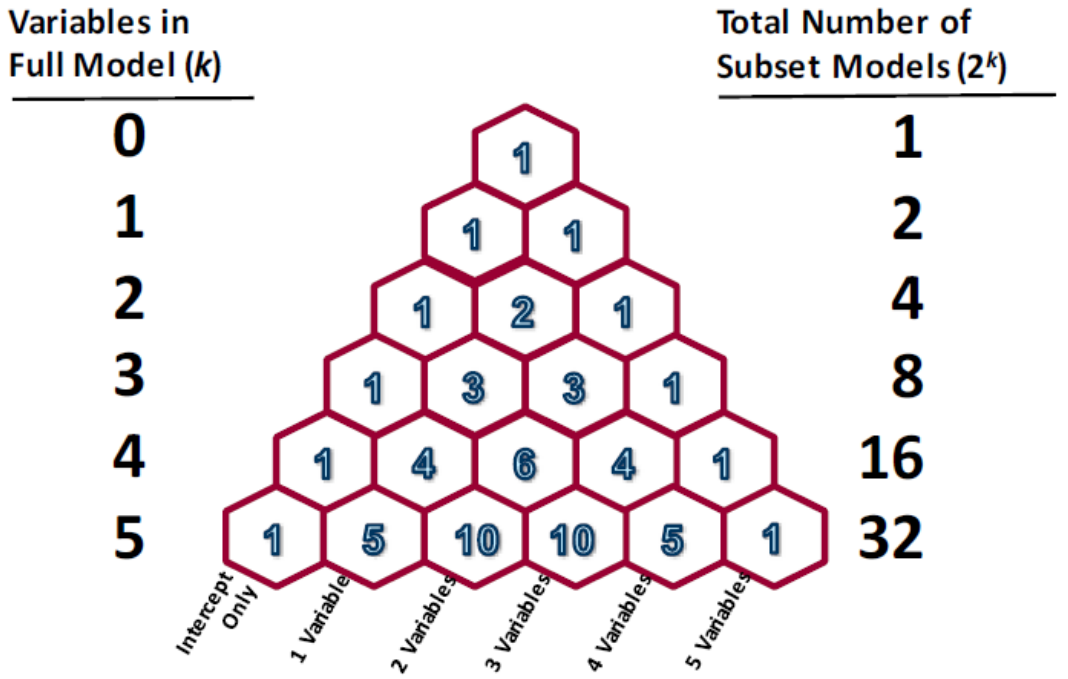
Which is the best input selection method?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Inputs Selection Methods

- The best way to find the optimal set of inputs for a regression model is to try the combination of every input variable.
- Unfortunately, the number of models to consider using this approach increases exponentially in the number of available inputs.



Do not edit
How to change the design



What is the primary goal of a predictive regression model?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Prediction (Predictive Analytics)

- The terms in the model, the values of their coefficients, and their statistical significance are of secondary importance.
- The focus is on producing a model that is the best at predicting future values of Y as a function of the Xs.
- The predicted value of Y is given by this formula:

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1} X_1 + \cdots + \widehat{\beta_k} X_k$$

Analytical or Explanatory Analysis (Diagnostic Analytics)

- The focus is on understanding the relationship between the dependent variable and the independent variables.
- Consequently, the statistical significance of the coefficients, the magnitudes and signs of the coefficients are important.

$$\hat{Y} = \widehat{\boldsymbol{\beta_0}} + \widehat{\boldsymbol{\beta_1}} X_1 + \cdots + \widehat{\boldsymbol{\beta_k}} X_k$$

Do not edit
How to change the design



Which of the following correctly matches the predictive model essential with their corresponding task in regression models?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Do not edit
How to change the design



Which of the following is not required for Generalised Linear Models?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Generalised Linear Models

The assumptions for Generalised Linear Models are,

- The observations are independent and random
- The response variable, Y, is from an exponential family
- The original response variable need not have a linear relationship with independent variables, but the transformed response variable is linearly related to independent variables.

Do not edit
How to change the design



The three elements of generalised linear models are:

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Three Elements of Generalised Linear Models

$$g(\mu_i) = \beta_0 + \sum_{i=1}^{k} \beta_i x_i$$

The three components of Generalised Linear Models

1. Systematic Component/ Linear Predictor
   - The linear combination of the predictors and the regression coefficients
2. Random Component
   - the probability distribution from an exponential family of the response variable.
3. Link Function, $g(\mu_i)$
   - specifies the function between the random and systematic components

Do not edit
How to change the design



Which of the following correctly matches a data issue with the appropriate solution?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Do not edit
How to change the design



A manager needs to identify the potential customers who will likely buy the new product. Which of the following models is the most suitable for him?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Do not edit
How to change the design



The odds ratio estimate for a numeric variable, salary is 1.56. it means that for each unit increment of the salary variable, the odds of the interest target level will (a) by (b).

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Do not edit
How to change the design



Which customer level will cause a quasi-complete separation issue to the binary logistic regression?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Quasi-Complete Separation

- Categorical variables can be included in regression models by introducing dummy variables for the levels.
- However, expanding categorical inputs into dummy variables can increase the dimension of the input space and produce many redundant and irrelevant inputs to the model, and the model may overfit the data.
- Other than that, quasi-complete separation may occur when the target variable is categorical.

Do not edit
How to change the design



The odds ratio estimate for a categorical variable, gender (male vs. female), is 1.85. This means that the odds of the interest target level for __ are __ times __ than the odds of the interest target level for __.

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Let us explore Logistic Regression Models using SAS Viya



MONASH University

Demo session

- Now, let's go through the demo to see how to build various types of Logistic regression models.
- The Week 6 Tutorial Demo Guide is uploaded in the Real-time section on Week 6 Moodle.
- Refer to the Week 6 Tutorial Demo Guide and the tutor's instructions.
- Then answer the following question.

MONASH University

Do not edit
How to change the design



Which selection method can add an effect and later remove it if it no longer satisfies the stay criterion?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Do not edit
How to change the design



Suppose the effect of customer tenure on the log odds of churn appears to be curved rather than constant across its range. Which Logistic Regression node option is most appropriate to test?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Do not edit
How to change the design



If the churn classification cutoff is reduced from 0.50 to 0.30, what changes would you expect, and what aspect of the fitted model should remain unchanged?

ⓘ The Slido app must be installed on every computer you're presenting from

slido

Q&A



MONASH University

Do not edit
How to change the design



Audience Q&A

ⓘ The Slido app must be installed on every computer you're presenting from

slido